Predicting microvascular barrier integrity using a 3D parametric model of the endothelial monolayer and the extracellular matrix

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Introduction

The endothelium consists of a monolayer of endothelial cells that line the interior surface of blood vessels. These cells form a dynamic structure, characterized by a constant process of intercellular adhesion unbinding and rebinding. This process creates gaps in the endothelium, allowing various substances and different cell types (including cancer cells) to enter and exit the blood vessels. Several studies have shown that mechanical and geometrical factors [1] influence gap formation in the endothelium and, consequently, its barrier integrity. These factors include the stiffness of the extracellular matrix surrounding the endothelial monolayer and its curvature. This study uses finite element simulation based on a parametric model to examine how these factors regulate gap opening in the endothelial monolayer.

Methods

A parametric geometry model was developed to facilitate the rapid generation of an endothelial monolayer (EM) and its surrounding extracellular matrix (ECM). This parametric model is able to reproduce two different and independent EM environments: in blood vessels (tubular ECM) and *in vitro* experiments (ECM with the shape of a microfluidic device). The intercellular stresses and gap formation on the monolayer were studied for different matrix stiffnesses and vessel diameters using this geometry and finite element simulation.

Linear elastic behavior was assumed for the cells and the substrate whereas a constitutive model based on a Catch Bond law [2] was used to simulate the cell-cell adhesions, taking into account both their dynamic behavior and stochastic nature. This model enables the simulation of the binding and unbinding processes of cell-cell junctions, facilitating understanding of how gaps in the EM open and close.

It is widely accepted [3] that calcium (Ca^{2+}) waves, which produce contraction in the cells, are one of the primary factors that induce changes in the barrier integrity of the EM. These phenomena were incorporated as loading conditions in the model, which was implemented in the commercial finite element software ABAQUS.

Results

The results of the simulations [4] indicate a correlation between matrices with greater stiffness and a higher percentage of gaps in the monolayer (Figure 1). This correlation remains consistent irrespective of substrate shape or vessel diameter. At a constant substrate stiffness, an increase in gap formation was observed as the vessel diameter decreased (Figure 1).

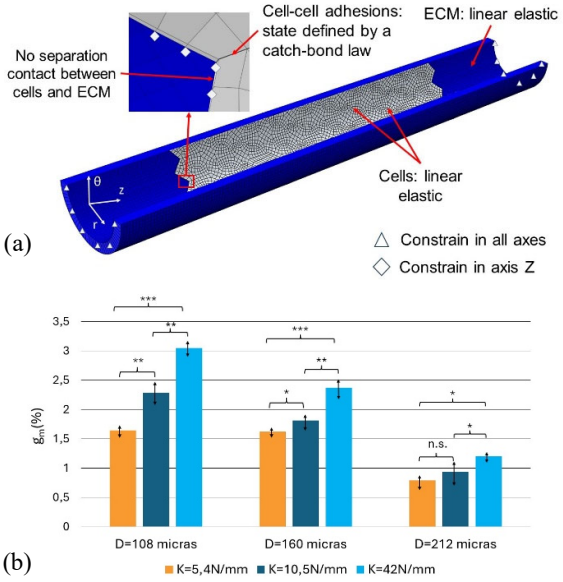


Figure 1: (a) Finite Element Model (tubular ECM). (b) Mean and standard deviation of the percentage of gaps (g_m) in the EM for different ECM stiffnesses and vessel diameters (tubular ECM). Statistical significance analysis: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

Conclusions

The findings of this study indicate that matrix stiffness and vessel curvature act in a synergistic manner to impair endothelial barrier integrity. Specifically, the combination of high substrate stiffness and reduced vessel diameter associated with tumor vascularity significantly increases the susceptibility to monolayer gap formation. These results are significant because of their potential implications for abnormal conditions, including cancer metastasis. This finding is consistent with experimental evidence [1].

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